CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 9: Triangles (Congruency in Triangles)

Topic-wise short notes with solved examples

The angle properties of a triangle, the conditions under which two triangles are congruent (SSS, SAS, ASA, AAS, RHS) and how to use congruent triangles to prove that sides or angles are equal.

1 Angles of a triangle

- The sum of the three angles of a triangle is $180°$.
- An **exterior angle** equals the sum of the two **interior opposite angles**, so it is greater than either of them.
- A triangle can have at most one right angle or one obtuse angle.

Example 1.1 In triangle ABC, $\angle A = 50°$ and the exterior angle at C is $120°$. Find $\angle B$ and $\angle C$.

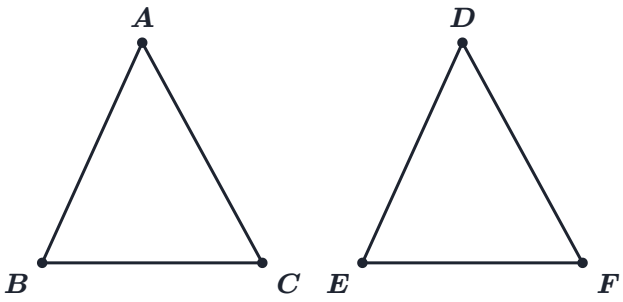
SOLUTION

$\angle B = 120° - 50° = 70°$ and $\angle C = 180° - 120° = 60°$

$\angle B = 70°, \angle C = 60°$

2 Congruent triangles

- Two triangles are **congruent** if they have exactly the same shape and size. One can be placed on the other so that it covers it exactly.
- We write $\triangle ABC \cong \triangle DEF$. The **order of the letters** shows which vertices correspond: A to D, B to E, C to F.
- **C.P.C.T.**: corresponding parts of congruent triangles are equal. So $AB = DE, BC = EF, CA = FD$, $\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$.



Triangle ABC is congruent to triangle DEF

3 Conditions for congruency

Condition	What must be equal
S.S.S.	Three sides of one triangle and the three sides of the other
S.A.S.	Two sides and the **included** angle
A.S.A.	Two angles and the **included** side
A.A.S.	Two angles and a corresponding side (not included)
R.H.S.	In right-angled triangles: the hypotenuse and one side

- **A.A.A. is not a condition**: equal angles give the same shape but not the same size.
- **S.S.A. is not a condition**: two sides and an angle that is not included do not fix the triangle (except in the R.H.S. case).

Example 3.1 In triangles ABC and PQR, $AB = PQ$, $\angle B = \angle Q$ and $BC = QR$. Are they congruent? If $\angle A = 65°$, find $\angle P$.

SOLUTION

Two sides and the included angle are equal: S.A.S.

$\triangle ABC \cong \triangle PQR$, **and** $\angle P = \angle A = 65°$ **(C.P.C.T.)**

Example 3.2 In triangles ABC and DEF, $\angle A = \angle D$, $\angle B = \angle E$ and $AB = EF$. Can we say they are congruent?

SOLUTION

AB lies between $\angle A$ and $\angle B$, but EF does not lie between $\angle D$ and $\angle E$. The equal sides are not corresponding.

No

4 Writing a proof

- State the two triangles.
- List three equal pairs, each with its **reason** (given, common side, vertically opposite angles, alternate angles, radii of the same circle, and so on).
- State the congruency with the correct condition and the vertices in corresponding order.
- Draw the conclusion by C.P.C.T.

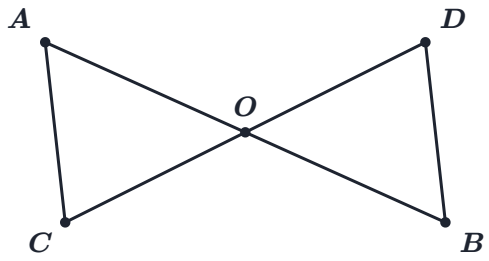
Example 4.1 AB and CD bisect each other at O. Prove that $AC = BD$ and that AC is parallel to BD.

SOLUTION

In $\triangle AOC$ and $\triangle BOD$: $AO = BO$ (given), $CO = DO$ (given), $\angle AOC = \angle BOD$ (vertically opposite).

So $\triangle AOC \cong \triangle BOD$ (S.A.S.), and $AC = BD$ (C.P.C.T.).

Also $\angle OAC = \angle OBD$ (C.P.C.T.). These are alternate angles, so $AC \parallel BD$.



AB and CD bisect each other at O

Example 4.2 In quadrilateral ABCD, $AB = AD$ and $CB = CD$. Prove that AC bisects $\angle BAD$.

SOLUTION

In $\triangle ABC$ and $\triangle ADC$: $AB = AD$, $CB = CD$, $AC = AC$ (common).

$\triangle ABC \cong \triangle ADC$ (S.S.S.)

$\angle BAC = \angle DAC$ (C.P.C.T.), so AC bisects $\angle BAD$

5 Results proved by congruency

- **Perpendicular bisector**: every point on the perpendicular bisector of a line segment is **equidistant from its two ends**. Conversely, a point equidistant from the ends lies on the perpendicular bisector.
- **Angle bisector**: every point on the bisector of an angle is **equidistant from the two arms**. Conversely, a point equidistant from the arms lies on the bisector.
- The perpendicular from the centre of a circle to a chord bisects the chord (R.H.S.).
- A diagonal of a parallelogram divides it into two congruent triangles (A.S.A.).

Example 5.1 P is a point on the bisector of $\angle ABC$. PM and PN are perpendiculars from P to BA and BC. Prove that $PM = PN$.

SOLUTION

In $\triangle PMB$ and $\triangle PNB$: $\angle PMB = \angle PNB = 90°$, $\angle PBM = \angle PBN$ (BP bisects the angle), $BP = BP$ (common).

$\triangle PMB \cong \triangle PNB$ (A.A.S.), so $PM = PN$

Example 5.2 In right triangles ABC and DEF, $\angle B = \angle E = 90°$, $AC = DF$ and $BC = EF$. If $AB = 5$ cm, find DE.

SOLUTION

Hypotenuse and one side are equal: R.H.S.

$\triangle ABC \cong \triangle DEF$, so $DE = AB = 5$ cm

6 Finding unknowns

- Once two triangles are known to be congruent, equate the corresponding parts to form equations.

Example 6.1 $\triangle ABC \cong \triangle PQR$. $AB = 3x + 1$, $PQ = 16$, $\angle C = 2y°$ and $\angle R = 70°$. Find x and y.

SOLUTION

$3x + 1 = 16$ and $2y = 70$

$x = 5$, $y = 35$

7 Common mistakes

- Writing the vertices in the wrong order, so that corresponding parts do not match.
- Using S.A.S. when the angle is not between the two sides.
- Claiming congruency from three equal angles.
- Using R.H.S. without a right angle, or with two sides that do not include the hypotenuse.
- Using C.P.C.T. before the triangles have been proved congruent.
- Leaving out the reason for a step.